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Experiment : 2

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int n;

    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        perror("Error creating destination file");
        close(source);
    }
}
```

```

    return 1;
}

while ((n = read(source, buffer, sizeof(buffer))) > 0)
{
    write(destination, buffer, n);
}

close(source);
close(destination);

printf("File copied successfully.\n");

return 0;
}

```

Output :

```

gssgvy@Gyanadeep:~$ cd ~/OSLab
gssgvy@Gyanadeep:~/OSLab$ pwd
/home/gssgvy/OSLab
gssgvy@Gyanadeep:~/OSLab$ mkdir -p Practical-02
gssgvy@Gyanadeep:~/OSLab$ cd Practical-02
gssgvy@Gyanadeep:~/OSLab/Practical-02$ pwd
/home/gssgvy/OSLab/Practical-02
gssgvy@Gyanadeep:~/OSLab/Practical-02$ gedit practical2.c &
[1] 560
gssgvy@Gyanadeep:~/OSLab/Practical-02$
** (gedit:560): WARNING **: 13:56:54.122: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:560): WARNING **: 13:56:54.122: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

[1]+  Done                  gedit practical2.c
gssgvy@Gyanadeep:~/OSLab/Practical-02$ gedit source.txt &
[1] 883
gssgvy@Gyanadeep:~/OSLab/Practical-02$
** (gedit:883): WARNING **: 13:57:28.524: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:883): WARNING **: 13:57:28.524: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

[1]+  Done                  gedit source.txt
gssgvy@Gyanadeep:~/OSLab/Practical-02$ gcc practical2.c -o practical2
gssgvy@Gyanadeep:~/OSLab/Practical-02$ ./practical2
File copied successfully.
gssgvy@Gyanadeep:~/OSLab/Practical-02$ cat destination.txt
Hello Operating Systems
This is Practical 02.
System calls are used for file operations.

gssgvy@Gyanadeep:~/OSLab/Practical-02$ strace --version
Command 'strace' not found, but can be installed with:
sudo apt install strace
gssgvy@Gyanadeep:~/OSLab/Practical-02$ sudo apt install strace
[sudo: authenticate] Password:
Installing:
  strace

Installing dependencies:
  libunwind8

```